Applications of Differential Equations

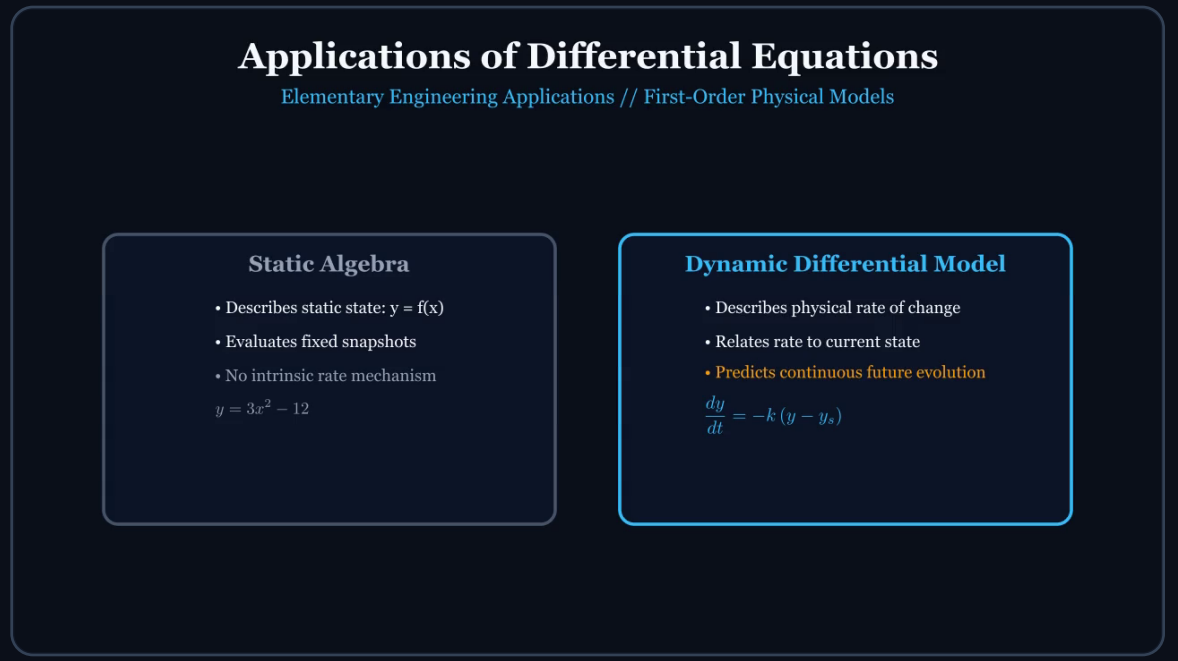
Elementary Applications of First-Order ODEs // Mathematical Modeling in Motion

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Dynamic Differential Modeling vs Static Snapshot Algebra
- Physical rates of change define future system trajectories
- Primary source: MATHalino Engineering Mathematics Review
- Universal workflow: Proportionality -> Separable ODE -> Particular Solution

LECTURER TELEPROMPTER & TEACHING HINTS

Introduce the lecture trajectory. Contrast static algebra with dynamical differential equations. Explain how derivatives embody physical conservation and balance laws.



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

The 4-Step Mathematical Modeling Blueprint

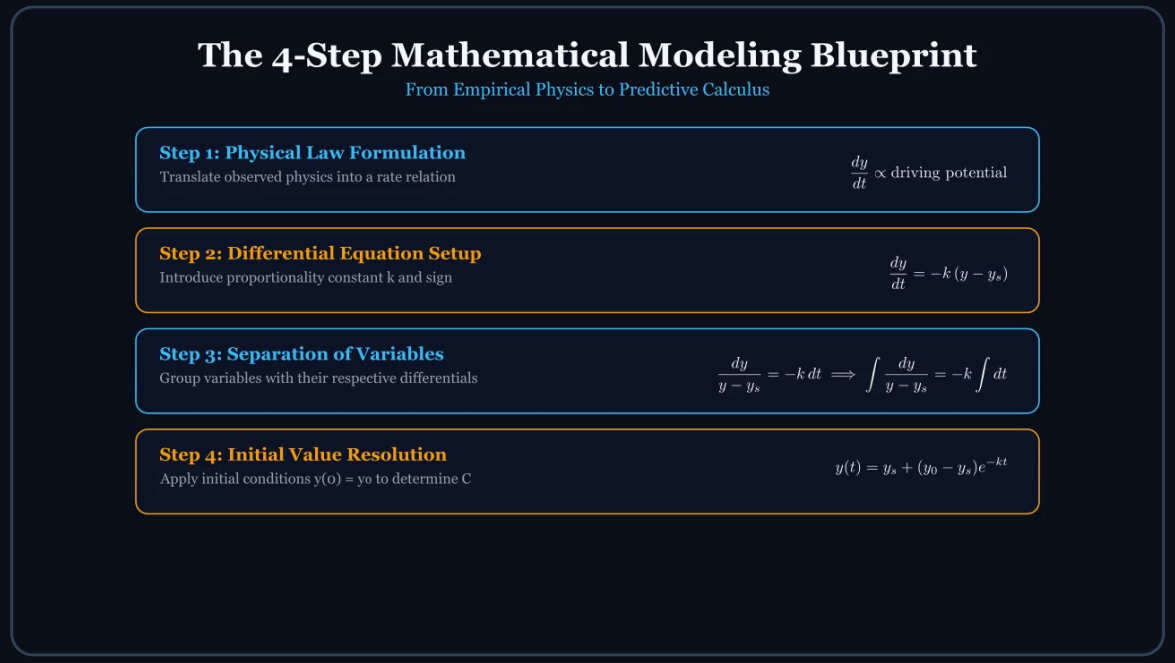
From Empirical Observation to Analytical Prediction

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Step 1: State the physical proportionality law (e.g. rate □ excess)
- Step 2: Formulate the differential equation with proportionality constant k
- Step 3: Separate variables and integrate to obtain general solution
- Step 4: Apply boundary and initial conditions to evaluate constant C

LECTURER TELEPROMPTER & TEACHING HINTS

Walk through the 4-step framework. Emphasize that all first-order rate processes follow this universal modeling blueprint.



□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Separation of Variables Mathematical Engine

Transforming Differential Statements into Integrated Solution Families

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Standard separable form: dy/dt = g(y) h(t)
- Algebraic isolation: dy / g(y) = h(t) dt across equality sign
- Dual integration: ∫ dy / g(y) = ∫ h(t) dt + C
- Explicit inversion: Solve for y(t) using exponentials and logarithms

LECTURER TELEPROMPTER & TEACHING HINTS

Demonstrate the mechanics of separation of variables. Highlight how Leibniz differentials dy and dt can be treated algebraically.

Separation of Variables Engine

Algorithmic Derivation for First-Order Autonomous & Factorable DEs

4. Explicit Analytical Solution Family

$$G(y) = H(t) + C \implies y(t) = G^{-1}(H(t) + C)$$

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Newton's Law of Cooling: Physical Intuition

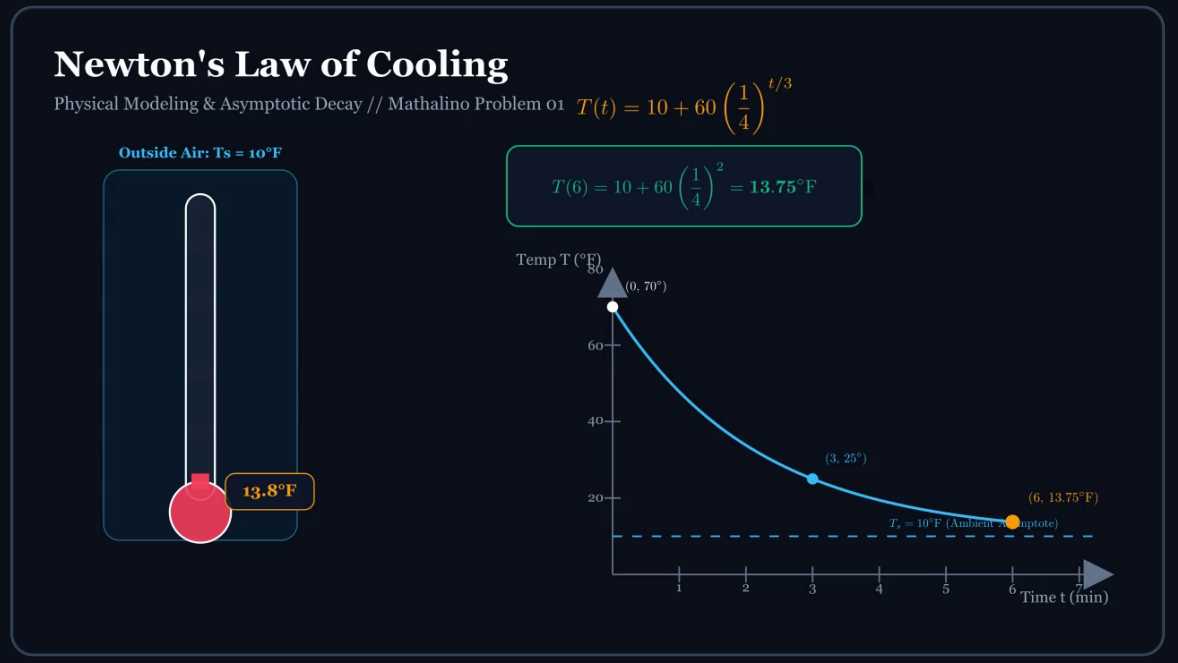
Thermal Driving Potential & Asymptotic Decay Toward Ambient Temperature

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Governing law: dT/dt = -k(T - Ts)
- Driving force is the temperature differential (T - Ts)
- Rate is fastest initially and decreases asymptotically
- Ambient medium Ts serves as an impassable horizontal asymptote

LECTURER TELEPROMPTER & TEACHING HINTS

Explain why dT/dt is proportional to (T - Ts). When T > Ts, the body cools (dT/dt < 0). Point out that temperature never drops below Ts under convective cooling alone.



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Newton's Law of Cooling: Analytical Derivation

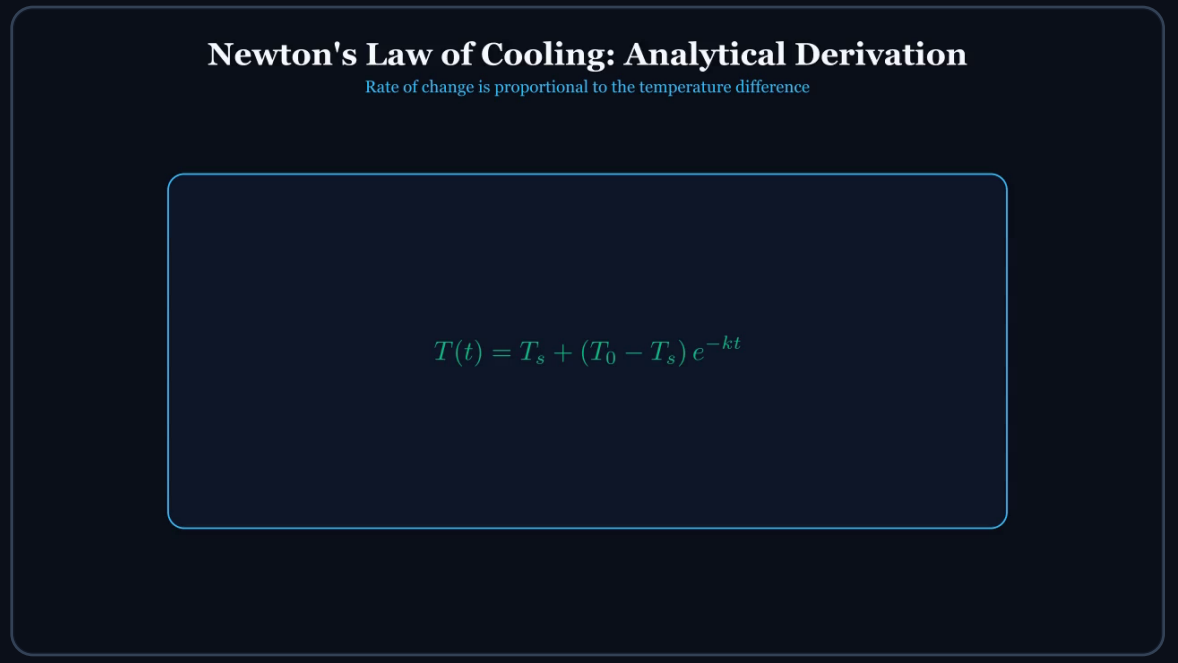
Continuous Equation Morphing from Rate Law to Explicit Solution

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Separated form: dT / (T - Ts) = -k dt
- Integrating both sides: ln(T - Ts) = -kt + ln C1
- Exponentiating: T - Ts = C1 e^(-kt)
- Initial condition: T(0) = T0 □ C1 = T0 - Ts
- Particular model: T(t) = Ts + (T0 - Ts) e^(-kt)

LECTURER TELEPROMPTER & TEACHING HINTS

Step through each line of the derivation. Emphasize how the initial condition evaluates constant C1 as initial temperature excess.



□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 01: Thermometer Reading After 6 Minutes

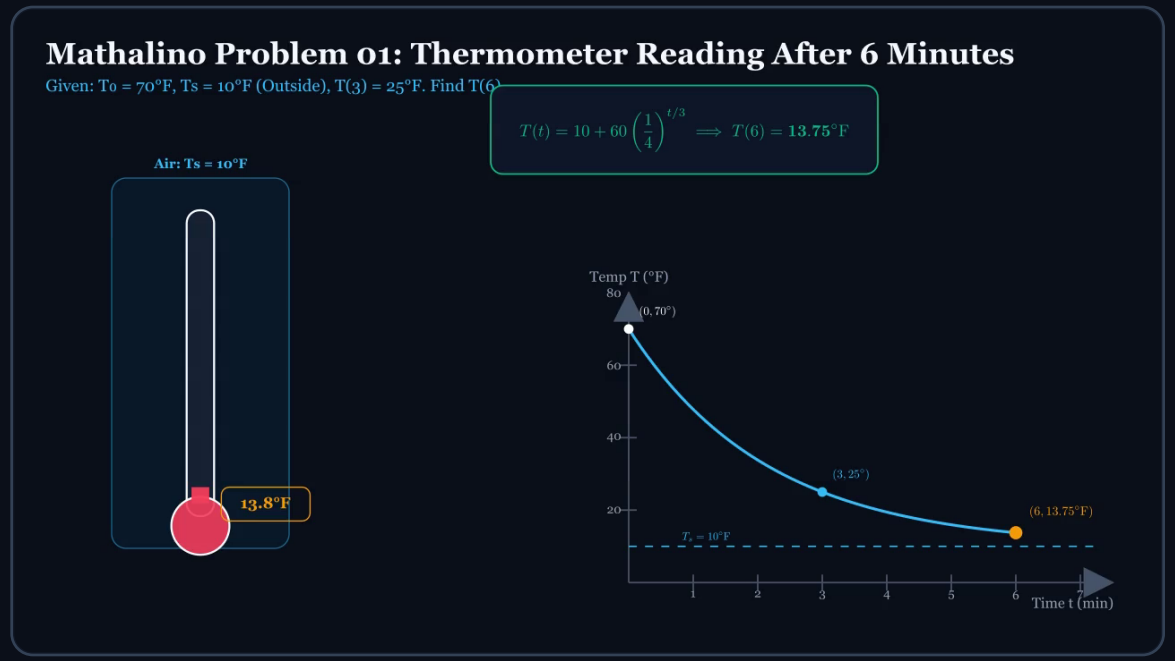
Given: Inside T□ = 70°F, Outside Ts = 10°F, T(3) = 25°F. Find T(6).

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Governing model: T(t) = 10 + 60 e^(-kt)
- At t = 3 min: 25 = 10 + 60 e^(-3k) □ e^(-3k) = 15/60 = 1/4
- Base model: T(t) = 10 + 60 (1/4)^(t/3)
- Evaluation at t = 6 min: T(6) = 10 + 60 (1/4)^2 = 10 + 3.75 = 13.75°F
- Independently verified against MATHalino engineering records

LECTURER TELEPROMPTER & TEACHING HINTS

State the problem conditions. Ask students to deduce e^(-3k) = 1/4 before revealing the ratio trick for t = 6 min.



□□ APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Problem 01 Resolution & Ratio Invariance

Constant Interval Scaling Principle in First-Order Linear Systems

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Interval 0 to 3 min: Excess drops from 60°F to 15°F (ratio = 1/4)
- Interval 3 to 6 min: Excess drops from 15°F to 3.75°F (ratio = 1/4)
- Final Temperature: T(6) = 10 + 3.75 = 13.75°F exactly
- Exponential decay scales identically over equal time steps

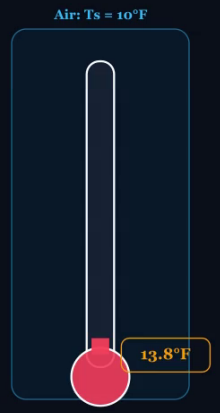
LECTURER TELEPROMPTER & TEACHING HINTS

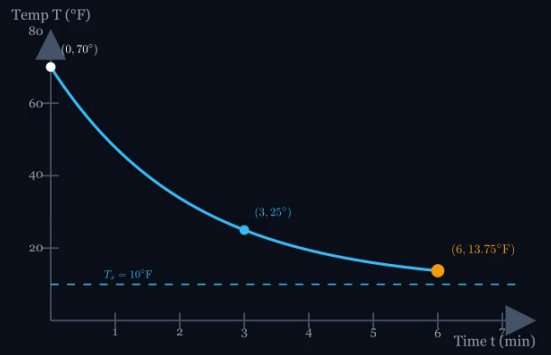
Use Apple Pencil on iPad to annotate the ratio arrows 60 -> 15 -> 3.75 on the temperature axis.

Mathalino Problem 01: Thermometer Reading After 6 Minutes

Given: $T_0 = 70°F$, $T_s = 10°F$ (Outside), $T(3) = 25°F$. Find $T(6)$

$$T(t) = 10 + 60\left(\frac{1}{4}\right)^{t/3} \implies T(6) = 13.75°\text{F}$$





APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 02: Time to Within Half Degree of Air

T□ = 75°F, Ts = 20°F, T(4) = 30°F. Target: T = 20.5°F. Find elapsed time t.

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Governing model: T(t) = 20 + 55 e^(-kt)
- At t = 4 min: 30 = 20 + 55 e^(-4k) □ e^(-4k) = 10/55 = 2/11
- Target condition: T(t) - 20 = 0.5°F □ T = 20.5°F
- Equation: 0.5 = 55 (2/11)^(t/4) □ 1/110 = (2/11)^(t/4)

LECTURER TELEPROMPTER & TEACHING HINTS

Explain why the target is 20.5°F (within half a degree of 20°F). Emphasize that reaching 20°F exactly would require infinite time.

Mathalino Problem 02: Time to Within Half Degree of Air

To = 75°F, Ts = 20°F, T(4) = 30°F. Target: T = 20.5°F. Find elapsed time t.

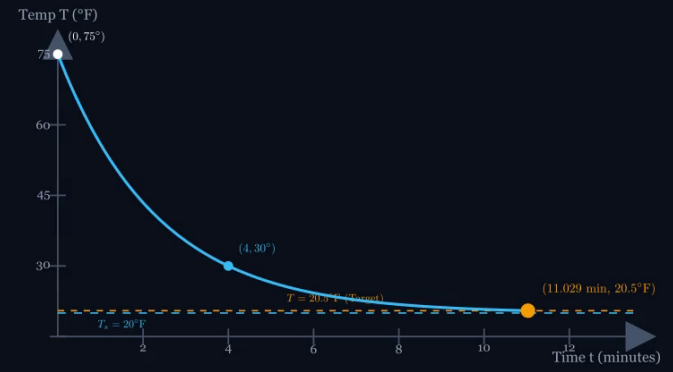
$$T(t) = 20 + 55\left(\frac{2}{11}\right)^{t/4}$$

$$20.5 = 20 + 55\left(\frac{2}{11}\right)^{t/4}$$

$$\frac{0.5}{55} = \frac{1}{110} = \left(\frac{2}{11}\right)^{t/4}$$

$$t = \frac{4\ln(1/110)}{\ln(2/11)}$$

$$t \approx 11.029 \text{ minutes}$$



□□ APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Problem 02 Logarithmic Resolution

Isolating Unknown Time Exponent via Natural Logarithms

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Logarithmic form: ln(1/110) = (t/4) ln(2/11)
- Exact solution: t = 4 ln(1/110) / ln(2/11)
- Numerical evaluation: t ≈ 11.029 minutes (11 min 1.7 sec)
- Independently verified against MATHalino official records

LECTURER TELEPROMPTER & TEACHING HINTS

Derive the logarithmic isolation step-by-step. Show how the signs cancel: ln(1/110) < 0 and ln(2/11) < 0.

Mathalino Problem 02: Time to Within Half Degree of Air

To = 75°F, Ts = 20°F, T(4) = 30°F. Target: T = 20.5°F. Find elapsed time t.

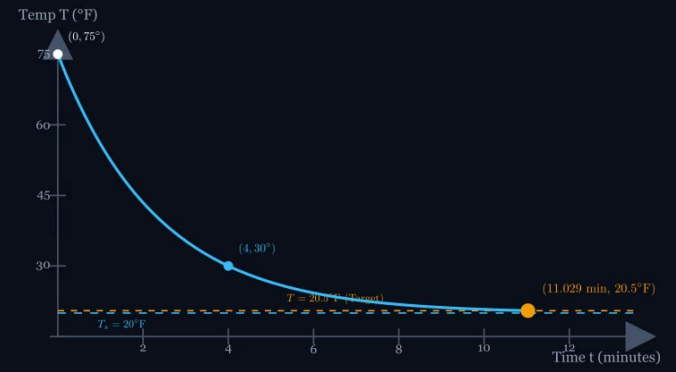
$$T(t) = 20 + 55\left(\frac{2}{11}\right)^{t/4}$$

$$20.5 = 20 + 55\left(\frac{2}{11}\right)^{t/4}$$

$$\frac{0.5}{55} = \frac{1}{110} = \left(\frac{2}{11}\right)^{t/4}$$

$$t = \frac{4\ln(1/110)}{\ln(2/11)}$$

$$t \approx 11.029 \text{ minutes}$$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Thermal Modeling Live Derivation Board

Open Annotation Canvas for Parameter Sensitivity & Boundary Variation

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Parameter sensitivity: Doubling k halves thermal relaxation time
- Varying Ts shifts the equilibrium asymptote up or down
- Inverse problem: Determining unknown heat transfer coefficient k
- Engineering application: Building insulation, electronics cooling

LECTURER TELEPROMPTER & TEACHING HINTS

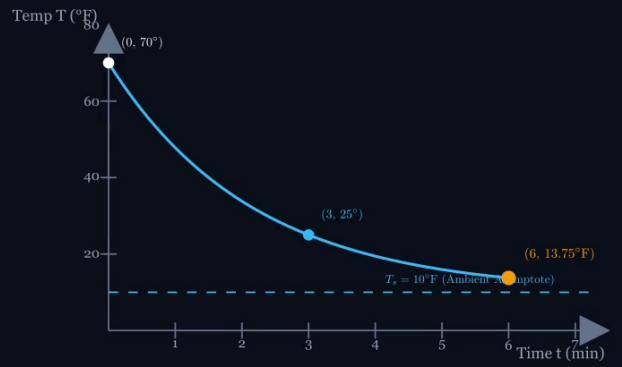
Dedicated whiteboard pause. Derive dT/dt for varying Ts, or sketch the effect of insulating the thermometer.

Newton's Law of Cooling

Physical Modeling & Asymptotic Decay // Mathalino Problem 01 $T(t) = 10 + 60\left(\frac{1}{4}\right)^{t/3}$



$T(6) = 10 + 60\left(\frac{1}{4}\right)^{2} = 13.75°\text{F}$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Chemical Conversion & Nuclear Decay Principles

First-Order Rate Law for Disappearing Reactants and Radioactive Isotopes

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Conversion rate is directly proportional to amount present: dx/dt = -kx
- Separated form: dx / x = -k dt □ ln(x) = -kt + ln(x0)
- Exponential decay function: x(t) = x0 e^(-kt)
- Unifies chemical kinetics, radioactive decay, and capacitor discharge

LECTURER TELEPROMPTER & TEACHING HINTS

Show how nuclear decay and first-order chemical reactions share the exact same differential equation dx/dt = -kx.

Simple Chemical Conversion & Radioactive Decay

Rate of decrease is proportional to the remaining amount present

$$x(t) = x_0\, e^{-kt}$$

□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Stochastic Atoms vs Continuum Calculus

How Billions of Discrete Quantum Events Manifest as Smooth Exponential Decay

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Each radioactive nucleus has a constant probability of decay per unit time
- At macroscale, Law of Large Numbers produces smooth continuum curve
- Half-life t_1/2 is strictly independent of starting quantity x0
- Universal relation: k * t_1/2 = ln(2) ≈ 0.69315

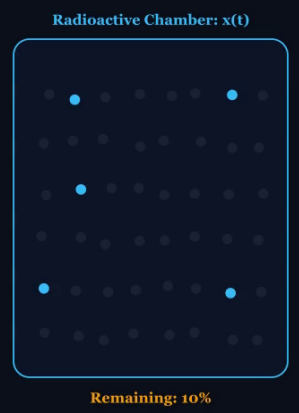
LECTURER TELEPROMPTER & TEACHING HINTS

Discuss the bridge between quantum randomness and macroscale deterministic calculus. Emphasize why half-life is constant regardless of sample mass.

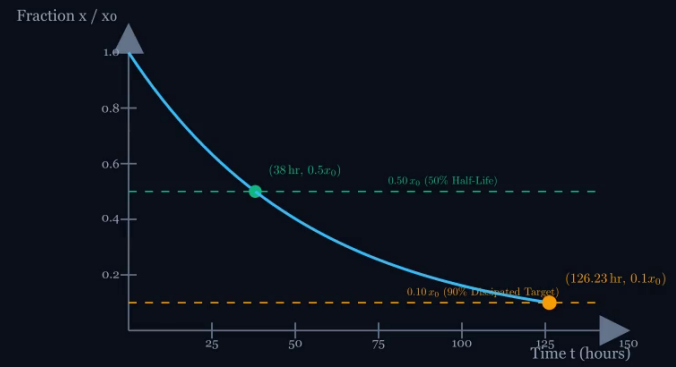
Simple Chemical Conversion & Radioactive Decay

First-Order Kinetics & Half-Life Thresholds // Mathalino Problem 02

$0.1x_0 = x_0(0.5)^{t/38} \implies 0.1 = 0.5^{t/38}$



$$t = \frac{38 \ln 0.1}{\ln 0.5} \approx 126.23 \text{ hours}$$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 01: Half-Life of Decomposing Radium

Given: 1.1% of radium decomposes in 25 years. Find half-life t_1/2.

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Remaining fraction at t = 25 yr: x(25) = (1 - 0.011) x0 = 0.989 x0
- Model: 0.989 x0 = x0 e^(-25k) □ e^(-25k) = 0.989
- Decay constant: k = -ln(0.989) / 25 ≈ 0.00044234 yr^(-1)
- Half-life equation: 0.5 x0 = x0 e^(-k t_1/2) □ t_1/2 = ln(2) / k

LECTURER TELEPROMPTER & TEACHING HINTS

Highlight that 1.1% decomposed means 98.9% remains. Students often mistakenly write x = 0.011x0.

Mathalino Decay 01: Radium Half-Life Determination

Given: In 25 yrs, 1.1% decomposes (x(25) = 0.989 x0). Find half-life (x = 0.5 x0).

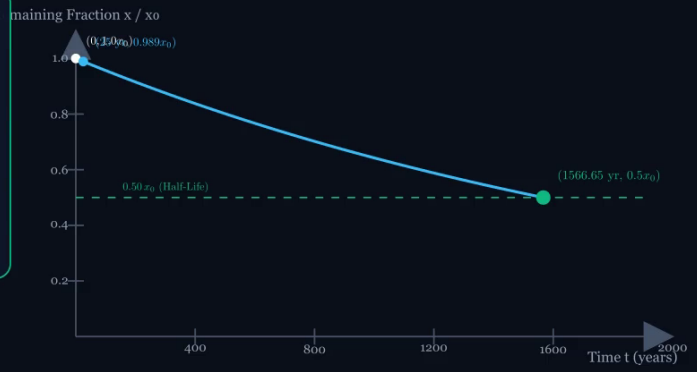
$$x(t) = x_0(0.989)^{t/25}$$

$$0.5x_0 = x_0(0.989)^{t/25}$$

$$0.5^{25} = 0.989^{t}$$

$$t = \frac{25 \ln 0.5}{\ln 0.989}$$

$$\mathbf{t \approx 1566.65}\ \text{years}$$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Problem 01: Century-Scale Resolution

Logarithmic Calculation of Radium Half-Life

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Combined formulation: t_1/2 = 25 ln(0.5) / ln(0.989)
- Calculation: 25 * (-0.693147) / (-0.011061) ≈ 1,566.65 years
- Literature comparison: Historical Radium-226 half-life ≈ 1,600 yr
- Verified against MATHalino official differential equations review

LECTURER TELEPROMPTER & TEACHING HINTS

Perform the division carefully. Note that ancient and geological dating rely on this exact differential calculation.

Mathalino Decay 01: Radium Half-Life Determination

Given: In 25 yrs, 1.1% decomposes (x(25) = 0.989 x0). Find half-life (x = 0.5 x0).

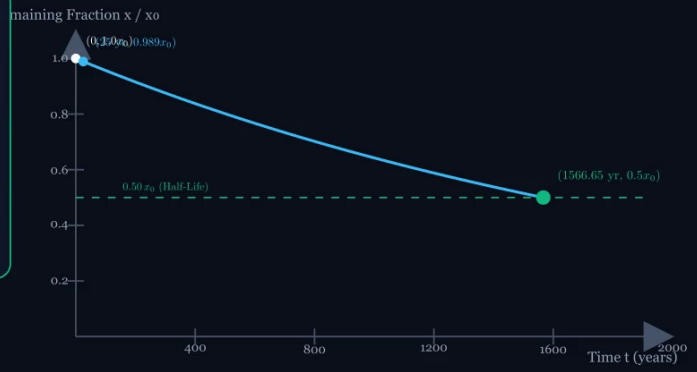
$$x(t) = x_0(0.989)^{t/25}$$

$$0.5x_0 = x_0(0.989)^{t/25}$$

$$0.5^{25} = 0.989^{t}$$

$$t = \frac{25 \ln 0.5}{\ln 0.989}$$

$$\mathbf{t \approx 1566.65 \text{ years}}$$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Mathalino Problem 02: Time to Dissipate 90% of Substance

Given: Half-life t_1/2 = 38 hours. Find time for 90% to dissipate.

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Half-life model: x(t) = x0 (1/2)^(t / 38)
- Condition: 90% dissipated □ 10% remains: x(t) = 0.10 x0
- Equation: 0.10 x0 = x0 (0.5)^(t / 38) □ 0.10 = (0.5)^(t / 38)
- Logarithmic relation: ln(0.10) = (t / 38) ln(0.5)

LECTURER TELEPROMPTER & TEACHING HINTS

Emphasize 90% dissipation means x = 0.10 x0. Point out that each 38 hours halves the remaining amount: 100% -> 50% -> 25% -> 12.5% -> 6.25%.

Mathalino Decay 02: 90% Radioactivity Dissipation

Given: Half-life $t_{1/2} = 38$ hours. Find time for 90% to dissipate (10% remaining).

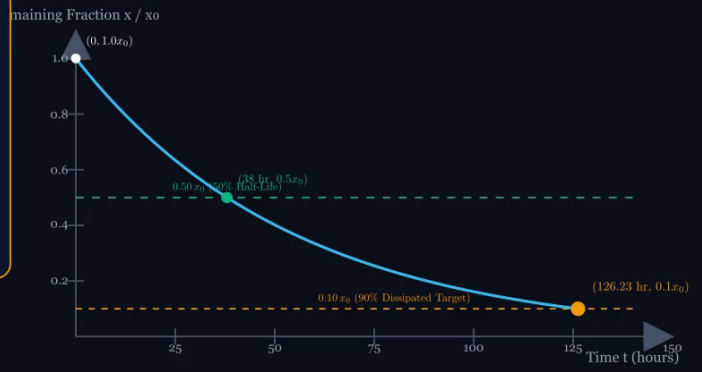
$$x(t) = x_0(0.5)^{t/38}$$

$$0.10x_0 = x_0(0.5)^{t/38}$$

$$0.10 = 0.5^{t/38} \implies 0.1^{38} = 0.5^{t}$$

$$t = \frac{38 \ln 0.10}{\ln 0.50}$$

$$t \approx 126.23 \text{ hours}$$



□□ **APPLE PENCIL LIVE WORKSPACE** *[GoodNotes / Notability / Keynote inking area]*

Problem 02: Threshold Resolution

Analytical Evaluation of 90% Dissipation Time

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Exact solution: t = 38 ln(0.10) / ln(0.5)
- Numerical calculation: 38 * (-2.302585) / (-0.693147) ≈ 126.23 hours
- Physical interpretation: 126.23 hr ≈ 5.26 days (between 3 and 4 half-lives)
- Verified against MATHalino engineering archive

LECTURER TELEPROMPTER & TEACHING HINTS

Show that 3 half-lives (114 hr) leaves 12.5% and 4 half-lives (152 hr) leaves 6.25%. Since 10% is between 12.5% and 6.25%, t must lie between 114 and 152 hr.

Mathalino Decay 02: 90% Radioactivity Dissipation

Given: Half-life $t_{1/2}$ = 38 hours. Find time for 90% to dissipate (10% remaining).

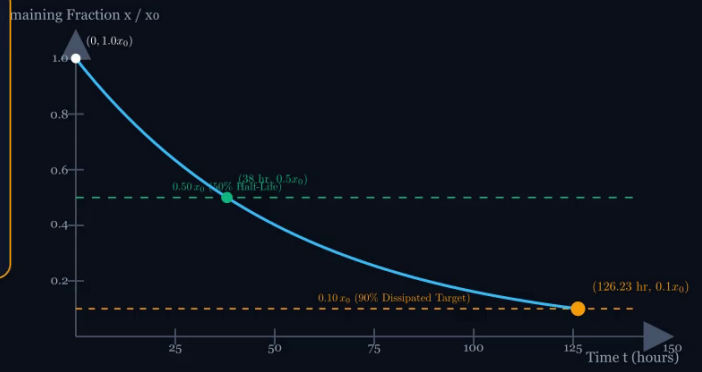
$$x(t) = x_0(0.5)^{t/38}$$

$$0.10x_0 = x_0(0.5)^{t/38}$$

$$0.10 = 0.5^{t/38} \implies 0.1^{38} = 0.5^{t}$$

$$t = \frac{38 \ln 0.10}{\ln 0.50}$$

$$t \approx 126.23 \text{ hours}$$



APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Extension: Velocity of Escape from Earth

Nonlinear Mechanics Application: Integrating Variable Gravitational Field

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

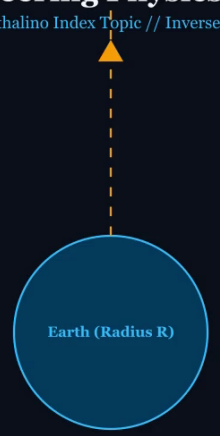
- Newton's 2nd Law + Universal Gravitation: m v (dv/dr) = -G M m / r^2
- Earth surface gravity relation: g = G M / R^2 □ G M = g R^2
- Separable differential equation: v dv = -g R^2 r^(-2) dr
- Boundary condition: At r = R, v = v_esc; as r -> ∞, v -> 0

LECTURER TELEPROMPTER & TEACHING HINTS

Introduce the acceleration substitution a = v dv/dr. Explain how the work-energy theorem naturally arises from integrating this first-order ODE.

Engineering Physics: Velocity of Escape from Earth

Primary Mathalino Index Topic // Inverse-Square Gravity Differential Equation



$$a = v\frac{dv}{dr} = -\frac{gR^2}{r^2}$$

$$v\,dv = -gR^2 r^{-2} dr$$

$$\int v\,dv = -gR^2 \int r^{-2} dr \implies \frac{1}{2}v^2 = \frac{gR^2}{r} + C$$

At $r = \infty$, $v \geq 0 \implies C \geq 0$

Minimum escape speed at surface ($r = R$):

$$v_{esc} = \sqrt{2gR} \approx 11.18 \text{ km/s } (6.95 \text{ mi/s})$$

□□ APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Escape Velocity Analytical Resolution

Definite Integration from Earth Radius R to Infinity

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

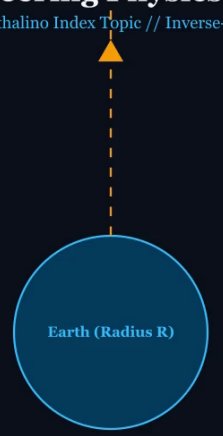
- Definite integral: ∫[v_esc -> 0] v dv = -g R^2 ∫[R -> ∞] r^(-2) dr
- Left side: [v^2 / 2] = -1/2 v_esc^2
- Right side: -g R^2 [-1/r] = -g R^2 (1/R) = -g R
- Escape velocity formula: v_esc = √(2 g R)
- Numerical evaluation: √(2 * 9.80665 * 6.371e6) ≈ 11.18 km/s (6.95 mi/s)

LECTURER TELEPROMPTER & TEACHING HINTS

Annotate the integral limits on the board. Emphasize that escape velocity is independent of the mass of the escaping projectile.

Engineering Physics: Velocity of Escape from Earth

Primary Mathalino Index Topic // Inverse-Square Gravity Differential Equation



$$a = v\frac{dv}{dr} = -\frac{gR^2}{r^2}$$

$$v\,dv = -gR^2 r^{-2} dr$$

$$\int v\,dv = -gR^2 \int r^{-2} dr \implies \frac{1}{2}v^2 = \frac{gR^2}{r} + C$$

At $r = \infty$, $v \geq 0 \implies C \geq 0$

Minimum escape speed at surface ($r = R$):

$$v_{esc} = \sqrt{2gR} \approx 11.18 \text{ km/s } (6.95 \text{ mi/s})$$

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Master Comparative Modeling Matrix

Structural Comparison of First-Order Applied Differential Equations

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Newton's Cooling: dT/dt = -k(T - Ts) [Bounded Exponential Relaxation]
- Radioactive Decay: dx/dt = -kx [Pure Exponential Decay to Zero]
- Chemical Conversion: dx/dt = k(a - x) [Progressive Mass Conversion]
- Escape Velocity: v dv/dr = -gR^2 / r^2 [Nonlinear Variable Field Mechanics]

LECTURER TELEPROMPTER & TEACHING HINTS

Synthesize the entire lecture. Compare and contrast how the restoring terms shape the asymptotic limits.

Comparative Modeling Matrix: 3 Core Applications

Common Differential Architecture Across Engineering & Physical Sciences

Newton's Cooling

Governing DE:

$$\frac{dT}{dt} = -k(T - T_s)$$

General Solution:

$$T(t) = T_s + (T_0 - T_s)e^{-kt}$$

Key Physics:

Thermal Equilibrium

Radioactive Decay

Governing DE:

$$\frac{dx}{dt} = -kx$$

General Solution:

$$x(t) = x_0\, e^{-kt}$$

Key Physics:

Half-Life Threshold

Escape Velocity

Governing DE:

$$v\frac{dv}{dr} = -\frac{gR^2}{r^2}$$

General Solution:

$$v = \sqrt{2gR^2/r}$$

Key Physics:

Gravitational Escape

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

Engineering Takeaways & Practical Wisdom

Best Practices for Solving Applied First-Order Models in Engineering

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- 1. Always check units: decay constant k must have units [time]^(-1)
- 2. Check asymptotic limits: as t -> ∞, does the solution reach physical equilibrium?
- 3. Preserve analytical ratios: (1/4)^(t/3) avoids intermediate rounding errors
- 4. Boundary conditions: Count constants of integration to ensure sufficiency

LECTURER TELEPROMPTER & TEACHING HINTS

Summarize core engineering heuristics. Remind students to always verify dimensional consistency.

The 4-Step Mathematical Modeling Blueprint

From Empirical Physics to Predictive Calculus

Step 1: Physical Law Formulation
Translate observed physics into a rate relation

$$\frac{dy}{dt} \propto \text{driving potential}$$

Step 2: Differential Equation Setup
Introduce proportionality constant k and sign

$$\frac{dy}{dt} = -k\,(y - y_s)$$

Step 3: Separation of Variables
Group variables with their respective differentials

$$\frac{dy}{y - y_s} = -k\,dt \implies \int \frac{dy}{y - y_s} = -k \int dt$$

Step 4: Initial Value Resolution
Apply initial conditions y(0) = yo to determine C

$$y(t) = y_s + (y_0 - y_s)e^{-kt}$$

APPLE PENCIL LIVE WORKSPACE *[GoodNotes / Notability / Keynote inking area]*

References & Academic Acknowledgements

Primary Source: MATHalino Engineering Mathematics Collection

KEY CONCEPTUAL & MATHEMATICAL TAKEAWAYS

- Primary Source: MATHalino Differential Equations Review (mathalino.com)
- Newton's Law of Cooling Problem 01 & 02 by Engr. Romel Verterra
- Radioactive Decay & Dissipation Problems by Engr. Romel Verterra
- Velocity of Escape from Earth derivation adapted from MATHalino mechanics
- Animations rendered with Manim Community Edition v0.22.0

LECTURER TELEPROMPTER & TEACHING HINTS

Acknowledge MATHalino and the open-source mathematics community. Direct students to the online reviewer for additional practice problems.

Applications of Differential Equations

Elementary Engineering Applications // First-Order Physical Models

Static Algebra

- Describes static state: y = f(x)
- Evaluates fixed snapshots
- No intrinsic rate mechanism

$y = 3x^2 - 12$

Dynamic Differential Model

- Describes physical rate of change
- Relates rate to current state
- Predicts continuous future evolution

$\frac{dy}{dt} = -k\,(y - y_s)$

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